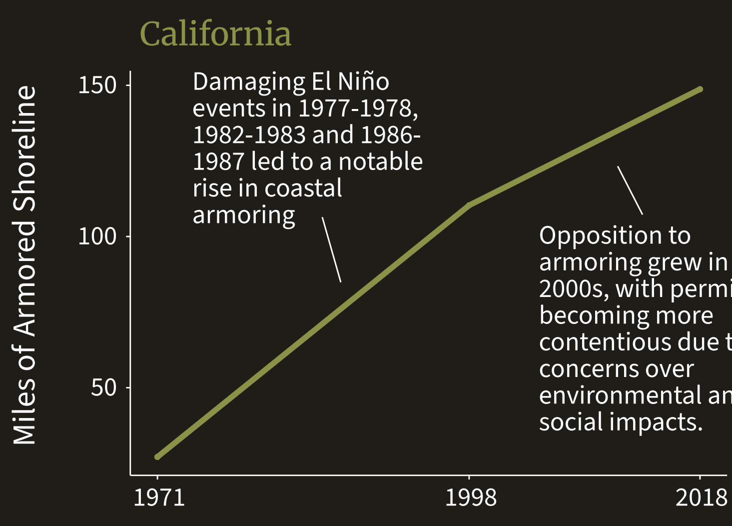
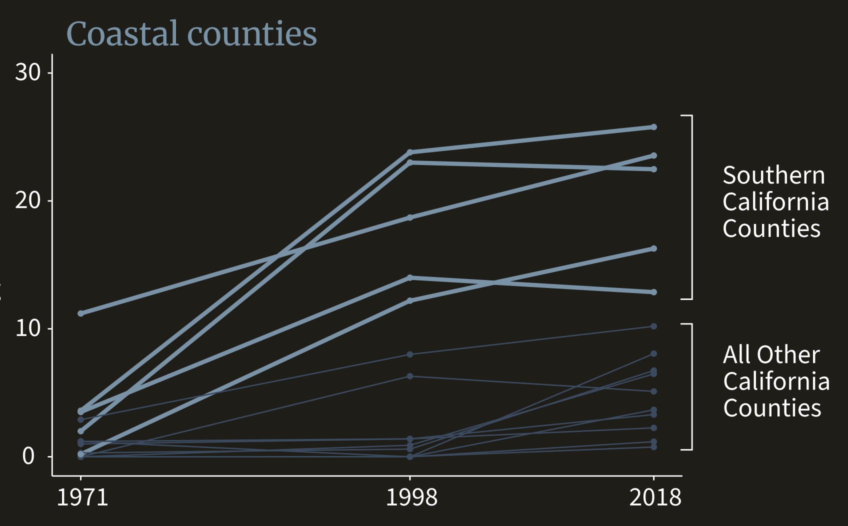
Breaking Down California's Shoreline Armoring

For decades, California's coastal communities have used armoring to prevent shoreline erosion and retreat. Now, as climate change accelerates sea level rise and intensifies storms, these structures threaten the resiliency of California's iconic beaches, bluffs, and coastal access.

13.3% of California's coastline is armored. That's **149** miles.





Seawall

Large concrete structures designed to protect the shoreline from tides, waves and storm surges.

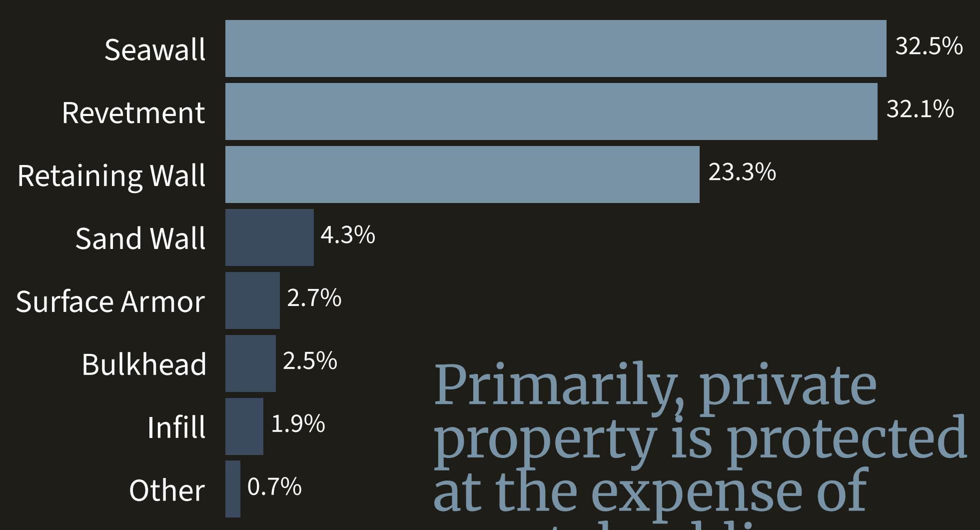
Retaining Wall

Small reinforcement walls primarily built to prevent cliff erosion.

Revetment

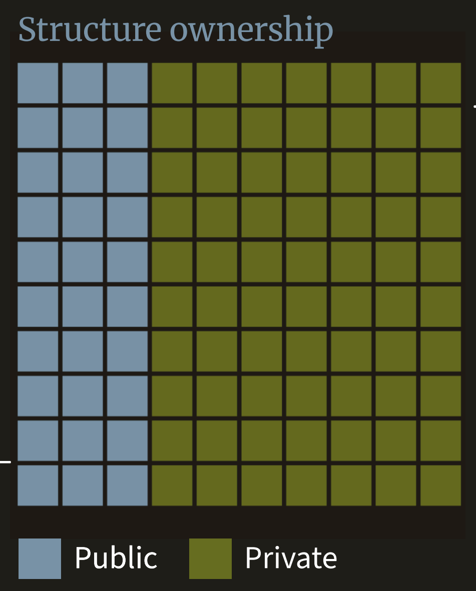
Sloped pile of rock or concrete primarily built to absorb wave energy and prevent erosion.

Southern California Counties have far more structures and miles of armoring than other coastal regions in the State.



Over 85% of armoring structures are seawalls, revetments, or retaining walls.

Primarily, private property is protected at the expense of coastal public spaces and critical habitats.



Most public structures are installed to protect local and state land, but a small proportion protect federal land.

Most private structures are installed to protect residential property, but a small proportion protect commercial property.

This raises long-standing concerns over the balance between private property rights and public interest in coastal access and conservation.

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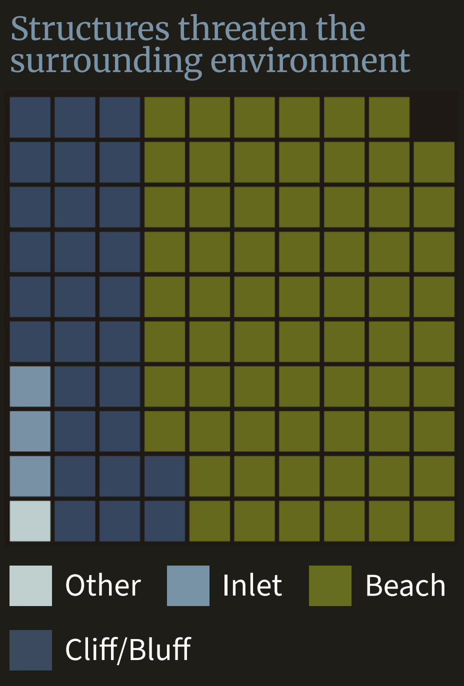
Shoreline armoring drives the coastal squeeze.

Shorelines naturally migrate landward with rising seas levels, but hard armoring structures like seawalls and revetments prevent this natural migration. This traps beaches and coastal habitats between rising waters and fixed barriers, resulting in significant loss of critical coastal habitat and public access areas.



Over 65% of armoring structures are located on beaches, threatening beach resilience in the face of climate change.

Solutions

In the long run, to fully resolve the issue of the coastal squeeze will require transformative adaptation of the areas where shoreline armoring, and the properties they protect, reside.

One relatively simple and cost-effective near-term solution is to remove broken structures. Failing structures no longer serve protective functions but continue to contribute to coastal squeeze. Removing these failed structures can help restore natural sediment transport processes and allow shorelines to migrate inland.

Data Sources: All armoring data with the exception of the timeseries data were sourced from the California Coastal Commission Open Data Portal, Coastal Armoring Dataset. Timeseries data were sourced from Griggs and Patsch., 2019. All photos were sourced from the California Coastal Records Project.

Griggs, G. and Patsch, K., 2019. The protection/hardening of California's coast: Times are changing. Journal of Coastal Research, 35(5), 1051–1061. Coconut Creek (Florida), ISSN 0749-0208.